

Regime Detection via Unsupervised Learning from Order Book and Volume Data

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1 Introduction

This document outlines a market regime analysis performed on financial time-series data using unsupervised machine learning approaches. The primary goal was to partition market dynamics into distinct operational regimes, which improves our comprehension of underlying market conditions and aids in formulating robust strategic decisions for trading operations.

2 Data Description

The dataset utilized for this study originates from two primary sources covering a period of four complete trading days:

- **Order Book Data:** Contains the top 20 levels for both bid and ask quotes, providing a granular view of market depth and available liquidity.
- **Trade Data:** Includes records of executed transactions, detailing trade prices, volumes, timestamps, and the side of the aggressor.

We engineered several custom features from this raw data. Furthermore, we introduced a novel attribute termed *trade direction*, categorized as follows:

- **1:** Indicates buy dominance
- **-1:** Indicates sell dominance
- **0:** Indicates a neutral or unclear direction

Prior to modeling, all extracted features were standardized and subsequently fed into the clustering algorithms.

3 Methodology

3.1 Feature Engineering

The engineered features are designed to reflect essential market microstructure properties, including volatility, bid-ask spread, order book imbalance, and the intensity of trades.

3.2 Dimensionality Reduction

Prior to applying clustering models, Principal Component Analysis (PCA) was utilized to lower the dimensionality and filter noise from the feature set. This step maintained the majority of the data's variance while significantly decreasing computational complexity.

3.3 Clustering Algorithms

The study evaluated two distinct clustering algorithms:

- K-Means Clustering
- Gaussian Mixture Models (GMM)

3.4 Evaluation Metrics

To evaluate model performance, the following metrics were computed:

- **Silhouette Score:** Quantifies the cohesion within clusters and separation between them (higher values denote better clustering).
- **Davies-Bouldin Index:** Evaluates the similarity among clusters (lower values indicate superior clustering performance).

4 Model Evaluation and Selection

Based on the results, the K-Means algorithm demonstrated superior performance over GMM on both evaluation metrics. Consequently, it was chosen for the in-depth market regime analysis.

Table 1: Performance Comparison of Clustering Algorithms

Algorithm	Silhouette Score	Davies-Bouldin Index
K-Means	0.4599	0.8914
GMM	0.0714	3.2176

5 Cluster Distribution

Table 2: Distribution of Data Points across K-Means Clusters

Cluster ID	Count
0	11,338
1	12,631
2	20,319

6 Regime Interpretation

Average market metrics were computed for each identified cluster:

Table 3: Average Market Characteristics by Cluster

Cluster	Volatility	Spread	Liquidity	Buy Ratio
0	0.0702	0.2971	-0.0175	0.2934
1	0.0899	0.7758	-0.0066	0.3758
2	0.0573	-0.6481	0.0139	0.2756

- **Cluster 0: Mean-Reverting and Moderately Volatile**

Exhibits moderate levels of volatility and bid-ask spread, pointing towards a range-bound market environment with slight liquidity constraints. This regime is typically favorable for mean-reversion trading strategies.

- **Cluster 1: Trending, Highly Volatile, and Illiquid**

Characterized by the highest volatility and widest spreads, coupled with low liquidity and an elevated buy ratio. These conditions indicate strong, aggressive directional price movements, making it an ideal environment for trend-following approaches.

- **Cluster 2: Stable, Highly Liquid, and Narrow Spread**

Features low volatility and tight spreads alongside high liquidity, reflecting a tranquil and highly efficient market state. This likely corresponds to periods of price consolidation or generally low market activity.

7 Visualizations

To facilitate analysis and ensure model validation, the following graphical representations were generated:

- Visualizations of cluster assignments derived from both K-Means and GMM algorithms.
- A time-series plot displaying the evolution of the detected regimes in relation to price and volatility metrics.
- A 2D t-SNE projection illustrating the separation and distribution of the identified clusters.

8 Conclusion

In conclusion, the study successfully delineated three distinct and meaningful market regimes through the application of K-Means clustering. These operational regimes accurately capture varying dynamics in volatility, liquidity, and trading behavior, providing valuable insights that can be effectively utilized for dynamic strategy selection and robust risk management.